NITRO-PURE:

A Passive Stove-Side NO_2 Reduction System

Design Proposal, Uniqueness Analysis, and Comprehensive Comparative Review

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ABSTRACT

Indoor nitrogen dioxide (NO_2) from residential gas stove combustion constitutes a significant and chronically underregulated public health hazard. Peak indoor NO_2 concentrations during cooking routinely exceed the U.S. Environmental Protection Agency (EPA) one-hour outdoor limit of 100 ppb, reaching 150–300 ppb in poorly ventilated kitchens. Epidemiological evidence links residential gas stove NO_2 exposure to approximately 200,000 childhood asthma cases and 19,000 premature deaths annually in the United States alone. Despite the availability of catalytic NO_2 reduction technology in automotive and industrial applications for over a decade, no device has ever been designed to intercept nitrogen oxides (NO_x) at the residential gas stove burner before dispersion into the indoor environment.

This paper presents Nitro-Pure: a passive, stove-mounted, catalytic filter concept for point-of-source NO_2 capture and reduction. The device employs a concentrically layered architecture comprising a Co-exchanged SSZ-13 zeolite mesh (Layer 1, 300–600°C) for methane-mediated selective catalytic reduction (CH_4-SCR), a Ca^{2+}-exchanged LTA zeolite mesh (Layer 2, 100–250°C) for residual NO_2 adsorption with zero NO by-product, and an activated carbon panel (Layer 3, <100°C) for polishing residual gases and volatile organic compounds. The system requires no electricity, no external chemical inputs, and no structural modification to the stove. The catalytic reductant is the stove's own combustion exhaust.

A systematic comparative analysis of all viable NO_2 reduction strategies confirms that no commercially available product, and no published research concept other than Nitro-Pure, addresses point-of-source catalytic NO_x interception at the residential kitchen burner. The Nitro-Pure architecture occupies a unique and previously vacant position in the indoor air quality solution landscape. Key open engineering challenges—including experimental validation of Co-SSZ-13 CH_4-SCR at residential stove conditions and thermal cycling durability of catalyst coatings—are identified and prioritised.

Keywords: indoor air quality; nitrogen dioxide; gas stove; selective catalytic reduction; Cu-zeolite; Co-SSZ-13; point-of-source filtration; residential air pollution; NO_2 abatement

1. INTRODUCTION

Gas and propane stoves are a significant but chronically underappreciated source of indoor air pollution. Unlike outdoor air quality, which is subject to regulatory monitoring and enforceable National Ambient Air Quality Standards (NAAQS) in most countries, indoor nitrogen dioxide (NO_2) from cooking combustion remains unregulated and largely unmonitored. Millions of households worldwide face daily exposures that would be considered legally impermissible in outdoor settings.

The scale of the health burden is substantial. In the United States, residential gas stove use is estimated to account for approximately 200,000 cases of childhood asthma and 19,000 premature deaths annually—a mortality burden equivalent to approximately 40% of that attributable to secondhand tobacco smoke. In India, where 62% of households rely on liquefied petroleum gas (LPG) as their primary cooking fuel, the population-level exposure risk is even greater.

Despite the availability of catalytic NO_x reduction technology in automotive and industrial contexts for over a decade, the residential kitchen burner has remained an entirely unaddressed application domain. No patent, no commercial product, and no published research proposal has described a catalytic NO_x filter designed for point-of-source deployment at a domestic gas stove.

This paper presents Nitro-Pure, a passive stove-side catalytic filter concept that addresses this gap. The paper is structured as follows: Section 2 characterises the NO_2 generation mechanism and associated health consequences. Section 3 introduces the Nitro-Pure design concept and catalytic mechanism. Section 4 presents a comparative benchmarking analysis against the three most scientifically proximate alternative technologies. Section 5 analyses the uniqueness of the approach and the structural reasons no prior work has addressed it. Section 6 provides a comprehensive comparative review of all viable NO_2 reduction strategies. Section 7 describes the updated dual-layer Nitro-Pure architecture incorporating findings from the comparative analysis. Section 8 identifies open engineering challenges and proposes a research agenda.

1.1 Scope and Objectives

The objectives of this paper are: (i) to characterise the public health problem of indoor NO_2 from residential gas stove combustion; (ii) to introduce and describe the Nitro-Pure catalytic filter design; (iii) to systematically compare Nitro-Pure against existing and emerging NO_2 reduction technologies; (iv) to establish the novelty and uniqueness of the point-of-source catalytic approach; and (v) to define the engineering research agenda required to advance the concept toward commercial readiness.

2. THE PROBLEM: NITROGEN DIOXIDE FROM RESIDENTIAL GAS STOVES

2.1 Mechanism of NO_2 Generation

When a residential gas or propane burner ignites, combustion occurs at flame temperatures exceeding 1,649°C. At these temperatures, the thermodynamic stability of atmospheric diatomic nitrogen (N_2) is disrupted: N_2 and O_2 react directly via the Zeldovich mechanism to form nitric oxide (NO), which subsequently oxidises in the cooler exhaust zone to produce nitrogen dioxide (NO_2). The relevant reactions are:

$$N_2 + O_2 \rightarrow 2\ NO \text{(Zeldovich, thermal pathway)}$$
$$2\ NO + O_2 \rightarrow 2\ NO_2 \text{(post-flame oxidation)}$$

Convection currents carry the NO_2-rich exhaust upward and laterally from the flame, dispersing pollutant into the kitchen environment. Under typical residential conditions, this dispersion reaches adjacent rooms within approximately one hour of stove use.

2.2 Observed Exposure Levels

The U.S. EPA sets a one-hour outdoor NO_2 safety limit of 100 parts per billion (ppb). Controlled kitchen studies consistently demonstrate that indoor concentrations during stove use substantially exceed this threshold:

- Boiling water on a gas burner raises indoor NO_2 to approximately 118 ppb
- Roasting meat drives concentrations above 300 ppb
- Routine daily stove use elevates background NO_2 by approximately 4 ppb, representing 75% of the World Health Organization (WHO) long-term annual exposure limit of 10 $\mu g/m^3$ (approximately 5.3 ppb)
- Peak concentrations immediately adjacent to the burner may be several orders of magnitude higher

The critical public health significance of these observations lies in the duration and proximity of exposure: the primary exposed individual—the cook—is located within the peak concentration zone for the entire cooking period.

2.3 Health Consequences

The health literature documents the following consequences of chronic and acute NO_2 exposure at residential cooking levels:

- Development of new asthma in previously healthy children and adults, and aggravation of pre-existing asthma
- Permanent obstructive and restrictive lung damage from chronic exposure, including chronic obstructive pulmonary disease (COPD) and chronic bronchitis
- An estimated 19,000 premature deaths per year in the United States, representing approximately 40% of the mortality burden attributable to secondhand tobacco smoke

- Approximately 200,000 childhood asthma cases in the US, with approximately 25% directly attributable to NO_2 from residential gas stove use
- Disproportionate burden in low-income and minority households, which have lower rates of range hood installation and window ventilation

2.4 Structural Barriers to Existing Solutions

The fundamental challenge of residential stove NO_2 is structural. NO_2 is generated at the point of combustion and disperses rapidly through convection. Room-scale solutions cannot prevent the acute peak exposure that occurs during the cooking event itself. Ducted range hood ventilation is highly effective but absent from most homes globally due to installation complexity and cost. The critical gap—a passive, point-of-source catalytic filter designed specifically for the kitchen burner environment—has never previously been addressed.

3. THE NITRO-PURE CONCEPT: A STOVE-SIDE CATALYTIC FILTER

3.1 Core Design Principle

Nitro-Pure is a passive, stove-mounted catalytic filter that intercepts NO_2-rich combustion exhaust directly at the gas burner before it disperses into the kitchen. The device consists of a perforated, catalyst-coated metal structure designed to sit concentrically around the burner grate, such that the upward convective exhaust flow necessarily passes through the catalytic media. No electricity, no maintenance beyond one annual component replacement, and no external chemical inputs are required.

3.2 Physical Configuration

The physical structure of the Nitro-Pure device comprises the following elements:

- Multiple concentric perforated stainless steel or heat-resistant mesh cylinders surrounding the burner at defined radii
- Pore walls and mesh wire surfaces coated with catalytic zeolite powder using a high-temperature-resistant inorganic adhesive binder
- Multi-layer concentrically arranged geometry to maximise cumulative contact area between exhaust gas and catalytic surfaces
- Footprint sized to fit within the existing standard stove grate dimensions, requiring no structural modification to the appliance

3.3 The Catalytic Mechanism of the Original Cu-Zeolite Design

Stage 1: Selective Adsorption

Chabazite (CHA) zeolite pores of approximately 3.8 Å match the molecular dimensions of NO_2, enabling selective capture. Copper(I) ions (Cu^+) at framework exchange sites attract NO_2 strongly via electrostatic interaction, given that NO_2 is a polar molecule with an unpaired valence electron. Non-polar gases—N_2, O_2, and CO_2—pass through the framework unimpeded.

Stage 2: Reduction at the Cu^+ Active Site

Adsorbed NO_2 gains an electron from Cu^+, being reduced to NO_2^- while Cu^+ is oxidised to Cu^{2+}. Thermal energy from the flame (200–600°C) cleaves the weakened N–O bond, allowing rearrangement into less toxic species that desorb from the zeolite lattice.

Stage 3: Self-Regenerating Redox Cycle

Cu^{2+} is re-reduced to the active Cu^+ state by carbon monoxide (CO) and unburned methane (CH_4) present in stove exhaust, restoring the catalyst. The system operates as a continuous, self-regenerating redox catalyst that is not consumed over normal operating lifetimes.

3.4 Zeolite Framework Selection: CHA (SSZ-13 / SAPO-34)

The chabazite (CHA) framework is selected for the following properties: (i) pore aperture of approximately 3.8 Å, closely matching the kinetic diameter of NO_2; (ii) exceptional thermal and hydrothermal stability across the temperature range encountered in stove exhaust (ambient to ~650°C); and (iii) robust support for the rapid Cu^+/Cu^{2+} redox cycling that drives the catalytic mechanism. The CHA zeolite family is the same framework used in all modern diesel vehicle SCR after-treatment catalysts—representing proven industrial technology adapted to a novel household scale application.

4. COMPARATIVE ANALYSIS: NITRO-PURE VS. KEY ALTERNATIVE TECHNOLOGIES

Table 1 benchmarks the Nitro-Pure approach directly against the three most scientifically relevant alternative technologies. All four represent the current frontier of NO_2 reduction science. The defining differentiator of Nitro-Pure is not its underlying chemistry alone—it is the specific application context: point-of-source catalytic capture at the kitchen burner.

Table 1. Comparative benchmarking of Nitro-Pure against key alternative NO_2 reduction technologies.

Criterion	Cu-Zeolite SCR(Nitro-Pure)	Ca^{2+}/Mn^{2+}-LTA(2025 Research)	Co-SSZ-13CH_4-SCR	HEPA +Activated Carbon
Point-of-source capture	Yes — at burner	No — room only	Yes — at burner	No — room only
Reductant required	CO/CH_4 from stove	None	CH_4 from stove	None
Ideal temp. range	200–550°C	≤25°C (ambient)	400–600°C	Ambient

Criterion	Cu-Zeolite SCR(Nitro-Pure)	Ca^{2+}/Mn^{2+}-LTA(2025 Research)	Co-SSZ-13CH_4-SCR	HEPA +Activated Carbon
Humidity tolerance	Moderate	Excellent (70% RH)	Moderate	Excellent
NO toxic by-product	Possible without NH_3	Zero	None (N_2 only)	N/A
NO_2 removal efficiency	70–95% (lab)	95–100% (lab)	60–80% (lab)	19–76% (clinical)
Deployment readiness	Lab/prototype	Lab only (2025)	Concept only	Commercial
Passive / no electricity	Yes	Yes	Yes	Yes
Kitchen-specific design	YES — first ever	No	No	No

SCR = Selective Catalytic Reduction; LTA = Linde Type A zeolite; HEPA = High-Efficiency Particulate Air filter.

4.1 Cu-Zeolite SCR (Nitro-Pure Original Design)

The Cu-zeolite SCR approach offers point-of-source capture—the critical differentiator—combined with a self-regenerating catalytic cycle that requires no external chemical input. The proven industrial heritage of CHA zeolite chemistry in automotive SCR provides a strong foundation for rapid development.

Key limitations requiring resolution include: (i) without a co-reductant such as NH_3, conversion to molecular nitrogen (N_2) rather than nitric oxide (NO) relies on the CH_4-SCR pathway, which requires Co^{2+} rather than Cu^{2+} catalysis; (ii) high-temperature steam (>600°C) can induce gradual zeolite dealumination over extended service life; and (iii) the original design produces NO rather than N_2 as the primary reduction product—remaining toxic at elevated concentrations. The updated dual-layer architecture described in Section 7 resolves both the product selectivity and the reductant pathway limitations.

4.2 Ca^{2+}/Mn^{2+}-LTA Zeolite (2025 Research)

The Ca^{2+}/Mn^{2+}-LTA system, published in 2025, achieves complete NO_2 capture with zero NO by-product under humid conditions—directly resolving two weaknesses of Cu-zeolite SCR. However, as an ambient-temperature adsorbent, placement proximal to the flame induces desorption and NO_2 release rather than capture. Its thermodynamically correct deployment is as a room-scale unit. It is therefore complementary to, rather than a replacement for, the point-of-source Nitro-Pure approach, and has been incorporated as Layer 2 of the updated architecture.

4.3 Co-SSZ-13 CH_4-SCR

The Co-SSZ-13 catalytic system represents the most compelling near-flame concept: a Co-exchanged chabazite catalyst that uses unburned methane from the stove as the reductant at 400–600°C, converting $NO_2 + CH_4 \rightarrow N_2 + CO_2 + H_2O$ with entirely benign products. This

mechanism has not been experimentally validated for residential kitchen conditions. It has been incorporated as Layer 1 of the updated Nitro-Pure architecture.

4.4 HEPA + Activated Carbon Room Purifiers

HEPA filtration combined with activated carbon represents the best commercially available solution today, offering 19–76% NO_2 reduction in clinical trials. These room-scale devices can reduce baseline indoor NO_2 concentration but are physically incompatible with near-flame placement and cannot intercept the acute concentration peak during cooking. They are best understood as complementary to a point-of-source approach, addressing different portions of the exposure profile.

5. UNIQUENESS OF NITRO-PURE: THE TECHNOLOGY GAP

5.1 Prior Art: What Exists and What Does Not

Selective catalytic reduction of NO_x is a mature industrial technology. Cu-zeolite SCR catalysts have been deployed in diesel vehicle exhaust after-treatment systems since the 2010s. Activated carbon air purifiers have been commercially available for decades. Zeolite adsorbents are widely applied in industrial gas separation. Yet none of these technologies has ever been applied, adapted, or proposed for the specific problem of residential gas stove kitchen NO_x abatement at the point of source.

> Although alternative technologies exist that can achieve equivalent or superior NO_2 filtration in laboratory and industrial settings, not one has been adapted or deployed to solve the specific problem of point-of-source indoor NO_2 from gas stove combustion. Nitro-Pure occupies a unique position: it is the first device concept designed to intercept NO_x at the kitchen burner using catalytic chemistry, before dispersion into the indoor environment.

5.2 Why Automotive SCR Was Not Adapted to Residential Kitchens

Automotive Cu-zeolite SCR systems address a superficially similar problem—NO_x from combustion—but under fundamentally different engineering constraints: highly concentrated and directed exhaust streams; integrated urea injection infrastructure; enclosed engine compartments providing controlled flow conditions; and large catalyst bed volumes with predictable residence times. Translating this technology to a passive kitchen device requires solving four problems that industrial SCR systems were never designed to address:

- Eliminating the NH_3/urea reductant supply chain entirely, substituting combustion exhaust CO and CH_4 as passive reductants
- Miniaturising the catalyst bed to fit within a standard burner grate footprint while maintaining adequate gas-catalyst contact time
- Engineering for daily thermal cycling from ambient to 500°C and back, unlike automotive systems that operate under quasi-steady-state thermal conditions

- Ensuring all materials are food-safe and that zero toxic by-products are emitted into a human breathing space

5.3 Why Room-Scale Purifiers Have Not Been Applied at Source

Activated carbon and zeolite-based room air purifiers operate on the design premise of dilution and long-residence-time adsorption across a large room volume. They are thermodynamically incompatible with near-flame placement: at temperatures above approximately 150°C, activated carbon risks ignition and zeolite adsorbents undergo thermal desorption, releasing previously captured pollutants. No commercial manufacturer has attempted a burner-proximal configuration, and no published research has proposed the concept.

5.4 The Core Innovation

Nitro-Pure's conceptual innovation is the recognition that a residential gas stove exhaust stream is chemically analogous to diesel engine exhaust: it contains NO_x, CO, and unburned hydrocarbons at elevated temperature. The same catalytic principles that solve diesel NO_x can therefore be adapted—passively and without the automotive urea infrastructure—to solve kitchen NO_x. The stove's own waste gases become the reductant. The stove's own heat becomes the thermal activator. No additional infrastructure, consumables, or energy inputs are required beyond annual activated carbon replacement.

5.5 Competitive Positioning Summary

The following properties, taken in combination, define the uniqueness of the Nitro-Pure concept in both the patent landscape and the published literature:

- First catalytic, point-of-source NO_x filter concept designed for a residential gas stove burner
- First application of automotive SCR catalyst chemistry to residential indoor air quality
- First device to use the stove's own combustion by-products (CO, CH_4) as the catalytic reductant, requiring no external chemical input
- Passive operation requiring no electricity, sensors, or user intervention beyond scheduled maintenance
- Addresses the acute peak exposure problem that all room-scale alternatives are structurally incapable of solving

> No patent, commercial product, or peer-reviewed publication describes a catalytic NO_x converter designed for residential gas stove point-of-source filtration. Nitro-Pure is the first such concept.

6. COMPREHENSIVE COMPARATIVE REVIEW: ALL VIABLE NO_2 REDUCTION STRATEGIES

Table 2 presents a systematic comparison of every practically viable approach to reducing indoor NO_2 from residential gas stove combustion, grouped by mechanism category. Effectiveness estimates are drawn from peer-reviewed clinical trials and controlled laboratory studies. The analysis establishes the landscape of available solutions, their practical limitations, and the specific gap that Nitro-Pure fills.

Table 2. Comprehensive comparative analysis of indoor NO_2 reduction strategies for residential gas stove applications.

Solution	NO_2 Reduction	Source Capture?	Passive?	Kitchen-Viable?	Status
SOURCE ELIMINATION					
Electric/induction stove replacement	~51–56% daily avg.	Yes — eliminates source	Yes	Yes — most effective	Commercial
LPG → PNG (piped natural gas) switch	Modest reduction	Partial	Yes	Limited — combustion remains	Infrastructure-dependent
VENTILATION					
Ducted range hood (exterior vented)	80–93% peak	Yes — removes exhaust	Fan required	Yes — highly effective	Commercial
Recirculating range hood (no duct)	5–18% (negligible)	No — recirculates	Yes	Inadequate alone	Commercial (misleading)
Open windows / natural ventilation	Variable: 20–60%	Partial	Yes	Partial — climate-dependent	Behavioural
ROOM-SCALE AIR PURIFICATION					
HEPA + activated carbon (e.g. Austin Air HM400)	19–76% avg.	No — room dilution only	Yes	Yes — best available today	Commercial
HEPA alone (no carbon)	<5% for NO_2	No	Yes	No — particles only	Commercial
Ozone / plasma generators	30–50% NO_2	No	Yes	No — ozone is toxic	Not recommended
Ca^{2+}/Mn^{2+}-LTA zeolite adsorbent (room unit)	95–100% (lab)	No — room only	Yes	Potentially — no product yet	Lab (2025)
MOF-based adsorbent purifier	80–95% (lab)	No	Yes	Not yet — humidity instability	Lab/research
TiO_2 UV photocatalysis	70–95% (UV lab)	No	Needs UV light	Limited — requires UV source	Pilot
POINT-OF-SOURCE CATALYTIC (AT BURNER)					

Solution	NO_2 Reduction	Source Capture?	Passive?	Kitchen-Viable?	Status
Cu-Zeolite SCR mesh (Nitro-Pure original)	70–95% est.	Yes — at burner	Yes	**YES — first kitchen design**	Prototype
Co-SSZ-13 CH_4-SCR (Nitro-Pure updated Layer 1)	60–80% est.	Yes — at burner	Yes — uses stove CH_4	Yes — if validated	Research concept
Catalytic converter-style inline hood insert	Est. 50–70%	Yes — captures exhaust	No — needs fan	Potentially — no product	Concept
BEHAVIOURAL / LOW-COST MEASURES					
Back burner use + fan direction	10–30% exposure reduction	No	Yes	Supplementary only	Behavioural
Cooking duration reduction	Proportional to time	No	Yes	Supplementary only	Behavioural

Highlighted row indicates the Nitro-Pure concept. All reduction figures are approximate; see references for primary data.

6.1 Key Findings

The Source Elimination–Room Purification Divide

Every commercially available solution occupies one of two categories: source elimination (removal of the gas stove) or room-scale dilution (treating air after dispersion). Source elimination is the most effective strategy by margin, but requires capital expenditure and infrastructure modification unavailable to renters, low-income households, and the vast majority of households in India and other developing markets. Room-scale dilution is accessible and widely deployed but cannot prevent the acute cooking-time exposure spike.

The Unaddressed Gap

No currently commercially available or commercially deployed product intercepts NO_x at the point of generation and converts it catalytically to harmless compounds before dispersion. This gap applies universally to HEPA purifiers, recirculating range hoods, ventilation fans, and every other solution evaluated in Table 2. Nitro-Pure is the only concept that addresses this gap directly.

Complementarity of Approaches

The solutions evaluated in Table 2 are not mutually exclusive. The optimal household exposure reduction strategy is layered: (i) Nitro-Pure at the burner for peak-exposure reduction at source; (ii) a ducted range hood where installation is feasible for bulk exhaust removal; and (iii) a room-scale activated carbon purifier for background concentration management. Each layer addresses distinct failure modes of the others.

Emerging Laboratory Solutions

The 2025 Ca^{2+}/Mn^{2+}-LTA zeolite and MOF-based adsorbent technologies demonstrate superior laboratory NO_2 capture performance but face 5–10 year product development timelines. Their thermal sensitivity near heat sources constrains their deployment to room-scale applications. They do not compete with Nitro-Pure; they represent complementary technologies at different stages of the solution stack.

> The comprehensive analysis confirms: Nitro-Pure is the only proposed solution that is simultaneously passive, point-of-source, catalytic, self-regenerating, and designed for the specific thermal and chemical environment of a residential gas stove burner.

7. UPDATED DESIGN: THE NITRO-PURE DUAL-LAYER ARCHITECTURE

The comparative analysis establishes that the original Cu-zeolite SCR design is sound in its point-of-source approach and self-regenerating redox cycle, but two design improvements are warranted: (i) the primary catalytic product should be N_2 rather than NO, requiring a catalyst and mechanism that achieves complete reduction; and (ii) a multi-zone architecture can deploy the optimal catalytic chemistry at each distinct temperature zone within the burner thermal field.

7.1 Exploitation of the Burner Thermal Gradient

The temperature field surrounding a residential gas burner is spatially heterogeneous. Flame-proximal inner zones reach 400–600°C; intermediate mid-zones 150–350°C; and outer far-field regions approach room temperature. Each thermal zone is chemically suited to a different catalytic or adsorptive material. The Nitro-Pure dual-layer architecture positions three concentric catalyst zones within a single filter body, each operating within its optimal temperature window.

Table 3. Nitro-Pure concentric tri-layer catalytic architecture.

Layer	Technology	Function
Layer 1 — Innermost (300–600°C zone)	Co-SSZ-13 zeolite mesh	CH_4-SCR: $NO_2 + CH_4 \rightarrow N_2 + CO_2 + H_2O$. High-temperature conversion to entirely benign products.
Layer 2 — Middle (100–250°C zone)	Ca^{2+}-LTA zeolite mesh	Ambient adsorption of residual NO_2 with zero NO by-product. Captures slip from Layer 1 at low flame settings.
Layer 3 — Outermost (<100°C zone)	Activated carbon + zeolite panel	Gas-phase polishing: residual NO_2, VOCs, CO, benzene. Annual replacement only.

SCR = Selective Catalytic Reduction; LTA = Linde Type A zeolite framework.

7.2 Layer 1: Co-SSZ-13 Catalytic Mesh (300–600°C)

The innermost layer employs Co-exchanged SSZ-13 (Co-SSZ-13) rather than the Cu-SSZ-13 of the original design. Co^{2+} at framework exchange sites performs C–H bond activation of methane at temperatures above 400°C—a capability absent in Cu^{2+} catalysis. This enables genuine CH_4-SCR according to the stoichiometry:

$$4\ NO_2 + CH_4 \rightarrow 4\ NO + CO_2 + 2\ H_2O \rightarrow \text{(further): } 2\ N_2 + CO_2 + 2\ H_2O$$

The stove's own methane slip from incomplete combustion serves as the passive reductant. At high flame settings, this layer converts the majority of the NO_2 spike entirely to benign N_2, CO_2, and H_2O products.

7.3 Layer 2: Ca^{2+}-LTA Zeolite Mesh (100–250°C)

The intermediate layer employs Ca^{2+}-exchanged LTA zeolite, operating at 100–250°C where adsorption capacity is effective. This layer captures NO_2 that bypasses Layer 1—particularly at low flame settings where Layer 1 operates below its catalytic activation threshold—and retains it as stable nitrate species with zero NO by-product. This layer directly incorporates the findings of the 2025 Ca^{2+}/Mn^{2+}-LTA research into the practical device architecture, adapting a room-scale research discovery to point-of-source deployment by positioning it at a thermally appropriate radius from the burner.

7.4 Layer 3: Activated Carbon Panel (<100°C)

The outermost layer is granular activated carbon—the proven adsorptive medium used in commercial room air purifiers. Positioned at a radius where temperatures remain below 100°C, this layer provides: (i) polishing of any residual NO_2 breakthrough from Layers 1 and 2; and (ii) co-capture of volatile organic compounds (VOCs), benzene, and carbon monoxide co-emitted by gas combustion. Layer 3 is the sole consumable component, requiring annual replacement. All other system components are permanent and maintenance-free.

7.5 Regeneration and Passivity

The self-regenerating character of the complete architecture is as follows:

- Layer 1 regenerates continuously via the Co^{2+}/Co^{+} redox cycle sustained by atmospheric oxygen in the exhaust stream
- Layer 2 undergoes thermal regeneration during high-intensity cooking events, when increased grate temperature drives desorption of saturated adsorption sites
- Layer 3 is replaced annually; all other components have indefinite operational lifetimes under normal use conditions
- No electricity, no external chemicals, no sensors, and no user intervention are required beyond Layer 3 annual replacement

> The updated Nitro-Pure architecture converts NO_2 to N_2 (not NO) at the burner, using only heat and methane already present in stove exhaust, while introducing zero new chemicals into the kitchen environment.

8. OPEN ENGINEERING CHALLENGES AND RESEARCH AGENDA

Three primary engineering challenges must be addressed before Nitro-Pure can advance from concept to validated prototype. These represent the priority research agenda for the next phase of development.

8.1 Experimental Validation of Co-SSZ-13 CH_4-SCR at Residential Stove Conditions

The Co-SSZ-13 CH_4-SCR mechanism has been characterised in laboratory settings with controlled gas compositions and flow rates, but has not been validated under the specific conditions of residential stove exhaust: variable NO_2 concentrations (50–500 ppb), low methane partial pressures (ppm-level methane slip), variable exhaust flow rates correlated to burner setting, and the presence of water vapour, CO_2, and kitchen aerosols. This is the most urgent experimental gap. Bench-scale validation using simulated residential stove exhaust is the essential next step.

8.2 Thermal Cycling Durability of Catalyst Coatings

Long-term adhesion of Co-SSZ-13 and Ca^{2+}-LTA zeolite powder coatings to perforated stainless steel substrates must be characterised across the daily thermal cycling profile of residential stove use: cold start from ambient temperature, rapid heating to 400–600°C, sustained high-temperature operation during cooking, and cooling back to ambient. Standard automotive SCR catalysts are washcoated onto cordierite monolith substrates under precisely controlled industrial conditions; adaptation to thin powder coatings on steel mesh under daily cycling conditions represents a distinct materials engineering challenge.

8.3 Gas Residence Time and Geometry Optimisation

The concentric layer geometry must be optimised to ensure adequate gas residence time within each catalytic zone. Residence time is the critical parameter governing conversion efficiency in Layer 1: the CH_4-SCR reaction requires sufficient contact time between the exhaust gas and the Co-SSZ-13 surface at temperature. Computational fluid dynamics (CFD) modelling of the convective exhaust flow field around a representative burner geometry, combined with bench-scale flow measurements, is required to define the optimal layer radii, perforation density, and mesh wire diameter.

8.4 Additional Considerations for Future Work

Secondary research questions for subsequent phases include: (i) food-safe material qualification for all catalytic components and binder materials; (ii) characterisation of by-product emissions (CO_2, H_2O, trace species) from the catalytic reactions; (iii) performance variation across different residential gas stove designs and fuel compositions (natural gas vs. LPG vs. PNG); (iv) long-term catalyst stability and defined service life under accelerated ageing; and (v) techno-economic analysis

of manufactured cost at volume for target markets including India and comparable LPG-dependent economies.

9. CONCLUSIONS

This paper presents Nitro-Pure, the first catalytic point-of-source NO_2 reduction device concept designed for the residential gas stove burner. The following conclusions are supported by the analysis:

1. Indoor NO_2 from residential gas stove combustion constitutes a substantive and underregulated public health hazard, with population-level health impacts comparable to secondhand tobacco smoke exposure in the United States and with disproportionate burden in developing economies reliant on LPG cooking.
2. The fundamental gap in the current solution landscape is the complete absence of any commercially available or proposed device that intercepts NO_x catalytically at the point of generation, before dispersion into the indoor environment. This gap is structural, not technical: the catalytic chemistry required has existed in industrial applications for over a decade.
3. Nitro-Pure addresses this gap through a passive, concentrically layered, thermally zoned catalytic filter architecture that exploits the burner's own thermal field and combustion exhaust as the activation energy source and reductant, respectively. The system requires no electricity, no external chemicals, and no structural modification to the appliance.
4. Systematic comparative analysis across all viable NO_2 reduction strategies confirms that no currently available technology—source elimination, ventilation, room-scale purification, or emerging laboratory systems—addresses the specific combination of point-of-source capture, passivity, self-regeneration, and kitchen-specific thermal design that defines Nitro-Pure.
5. Three priority engineering challenges require experimental validation before prototype advancement: Co-SSZ-13 CH_4-SCR performance at residential stove conditions; thermal cycling durability of zeolite coatings on steel substrates; and gas residence time optimisation via CFD and bench-scale flow characterisation.

Nitro-Pure represents a previously unexplored but technically grounded approach to a significant public health problem. The convergence of mature catalytic chemistry, proven zeolite materials, and an unaddressed application domain creates an opportunity for meaningful impact in both developed and developing markets.

Author Contributions

[To be completed per journal requirements, e.g., CRediT taxonomy: Conceptualisation, X.X.; Methodology, X.X.; Writing—Original Draft, X.X.; Writing—Review & Editing, X.X.]

Funding

This research received no external funding. [Or: This research was funded by [Funder Name], grant number [XXX].]

Conflicts of Interest

The authors declare no conflicts of interest.

Ethical Statement

This paper presents a design concept and does not report results from human or animal subjects. No ethical approval was required.

Data Availability

No original experimental data are reported in this paper. All data cited are derived from published literature as referenced.

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